
Assignment 3

Rules

- The **deadline** is **23:59, May 07, 2025**.
- The weight of this assignment in the final grade is 20%.
- **Electronic submission:** Turn in solutions electronically via Blackboard. Be sure to submit your answers as one pdf file plus one ipynb script file for programming questions. Please name your solution files as `HW3_studentID_name.pdf` and `HW3_studentID_name.ipynb` (.py file are also acceptable).
- The complete and executable codes must be submitted. If you only fill in some of the results in your answer report for programming questions and do not submit the source code (.py or .ipynb files), you will receive **0 points** for the question.
- Note that **late submissions** will result in discounted scores: 0-48 hours $\rightarrow$ 50%, more hours $\rightarrow$ 0%.
- Answer the questions in English. Otherwise, you'll lose half of the points.
- Collaboration policy: You need to solve all questions independently and collaboration between students is **NOT** allowed.
- **Score: 100 points (50pts writing, 50pts coding)**
- **Important Note:** **Ensure the manual nature of the work**, any AI-assisted work will be queried for manual components. Solutions that are completely output by black-box models will not be scored.
- **Questions on assignment:** Start early and come to TA office hours with your questions on the assignments.

1 Written Problems (50 points)

1.1. Mixture of Bernoulli In lecture we introduced the Gaussian mixture model (GMM). In this question, however, we consider a mixture of Bernoulli distributions. In this case, our datapoints are denoted as m -dimensional vectors of binary values $\mathbf{x} \in \{0, 1\}^m$.

Let $\boldsymbol{\mu}_k$ denotes a mean vector, a single multivariate Bernoulli distribution can be formulated as

$$p(\mathbf{x}|\boldsymbol{\mu}_k) = \prod_{j=1}^m \mu_k[j]^{\mathbf{x}[j]} (1 - \mu_k[j])^{1-\mathbf{x}[j]},$$

where $\mu_k[j]$ and $\mathbf{x}[j]$ denotes the j -th value of $\boldsymbol{\mu}_k$ and $\mathbf{x}$, respectively. And we define a parameter set of mixture as $\Theta = \{\boldsymbol{\mu}_k, \pi_k | k = 1, \dots, K\}$

- (3 points) write a mixture of K multivariate Bernoulli distributions $p(\mathbf{x}|\Theta)$.
- (4 points) suppose we have dataset $X = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$ (this definition also works in the following problem), write an expression for log-likelihood $L(\Theta)$
- (6 points) Now, suppose that we divide the datapoints $\mathbf{x}$ into two parts $\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2]$ with $\mathbf{x}_1 \in \{0, 1\}^{m-d}$ and $\mathbf{x}_2 \in \{0, 1\}^d$. Please show that the conditional distribution $p(\mathbf{x}_1|\mathbf{x}_2; \Theta)$ is also a Bernoulli mixture distribution and provide expressions for the mixing coefficients and the component densities of this new mixture.

In the following question, we first introduce a binary latent variable $\mathbf{z}^{(i)}$ associated with each instance of $\mathbf{x}^{(i)}$. Unlike Gaussian mixture, here $\mathbf{z} = (z_1, \dots, z_K)^\top$ is a binary K -dimensional variable having a single component equal to 1, with all other components equal to 0. Let $Z = \{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(n)}\}$

- (6 points) What is $p(\mathbf{z}^{(i)}|\pi)$, $p(\mathbf{x}^{(i)}|\mathbf{z}^{(i)}, \Theta)$, and $p(Z, X|\Theta)$?
- (6 points) Let $\phi(z_k^{(i)}) = \mathbb{E}[z_k^{(i)}|\mathbf{x}^{(i)}, \Theta]$. Show that

$$\phi(z_k^{(i)}) = \frac{\pi_k \prod_{j=1}^m \mu_k[j]^{\mathbf{x}^{(i)}[j]} (1 - \mu_k[j])^{1 - \mathbf{x}^{(i)}[j]}}{\sum_k \pi_k \prod_{j=1}^m \mu_k[j]^{\mathbf{x}^{(i)}[j]} (1 - \mu_k[j])^{1 - \mathbf{x}^{(i)}[j]}}$$

- (4 points) Derive the E-step expression $\mathbb{E}[\ln p(Z, X|\Theta)|X, \Theta]$
- (6 points) Show that the values of $\tilde{\mu}_k$ and $\tilde{\pi}_k^*$ that maximize the E-step expression are

$$\tilde{\mu}_k = \frac{\sum_{i=1}^n \phi(z_k^{(i)}) \mathbf{x}^{(i)}}{N_k}, \quad \tilde{\pi}_k = \frac{N_k}{\sum_k N_k}$$

where $N_k := \sum_{i=1}^n \phi(z_k^{(i)})$. Remark: consider the constraint $\sum_{k=1}^K \pi_k - 1 = 0$

1.2. Confusion Matrix

		PREDICTED		Total
		Positive (1)	Negative (0)	
ACTUAL	Positive (1)	TP = 20	FN = 5	25
	Negative (0)	FP = 10	TN = 15	25
	Total	30	20	50

Figure 1: Two-class confusion matrix

- (4 points) See a case of two-class confusion matrix shown in Figure 1, calculate the accuracy, precision, recall, and F1-Score with the definition:

$$\text{F1score} = \frac{2}{\text{Recall}^{-1} + \text{Precision}^{-1}}$$

		PREDICTED classification					
		Classes	a	b	c	d	Total
ACTUAL classification	a	5	23	17	17	62	
	b	10	540	21	14	585	
	c	166	96	436	110	808	
	d	1	2	5	87	95	
	Total	182	661	479	228	1550	

Figure 2: Multi-class Confusion Matrix

- (2 points) Classification tasks in machine learning involving more than two classes are known by the name of "multi-class classification". See a case of multi-class confusion matrix shown in Figure 2. What are the TP, FP, FN, TN for class *b*?
- (9 points) Calculate the accuracy, macro average precision, macro average recall, and balanced accuracy of Figure 2 with the definition:

$$\text{MacroPrecision} = \frac{\sum_{k=1}^K TP_k / (TP_k + FP_k)}{K}$$

$$\text{MacroRecall} = \frac{\sum_{k=1}^K TP_k / (TP_k + FN_k)}{K}$$

$$\text{BalancedAccuracy} = \frac{\sum_{k=1}^K TP_k / (Total_{row_k})}{K}$$

where k denotes the class label. In this example, is the balance accuracy the same as the accuracy? Why?

2 Programming (50 points)

Important Notes

1. All instructions and questions are provided in the accompanying Jupyter notebook file **Assignment3_KMeans_GMM.ipynb**.
2. Please make sure to complete all the code and analysis cells. Specifically, fill in all sections marked with comments such as **## Write your code here ##**.
3. You are not required to copy your answers from the notebook into the written report. However, you must submit the completed notebook file as part of your assignment.
4. Do not include any datasets or generated data files in your submission.
5. Ensure that all code cells are executed and outputs (such as plots, printed results, and metrics) are visible in your final notebook. Incomplete or non-executed notebooks will receive point deductions.

2.1. (K-Means Clustering) In this section, you will implement the K-Means clustering algorithm from scratch. This includes initializing cluster centers, assigning points to clusters based on distance, updating centers iteratively, and evaluating clustering quality using the Silhouette Score. You will also visualize the clustering results and interpret the output.

2.2. (Gaussian Mixture Model with EM) In this section, you will implement a Gaussian Mixture Model (GMM) using the Expectation-Maximization (EM) algorithm. This involves computing the multivariate Gaussian density, implementing both the E-step and M-step, and fitting the model to the dataset. You will visualize the resulting clusters and evaluate model performance using the Silhouette Score.